TASFIA JANNAT TASFI

Natural Language Processing | LLM Evaluation | Trustworthy AI

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Research Profile

Computer Science and Engineering graduate with research experience in **language-model evaluation, multimodal credibility analysis, and trustworthy NLP**. Co-authored studies of cross-lingual reasoning failures in instruction-tuned LLMs and semantic deepfake verification using multimodal language models. My thesis combines transformer-based medical misinformation detection, uncertainty calibration, and human evaluation of AI-assisted decisions. I aim to investigate **knowledge-aware LLM reasoning, factuality, and multilingual robustness**.

Research Interests: LLM Reasoning & Evaluation; Knowledge-Aware NLP; Factuality & Misinformation; Multilingual NLP; Multimodal Verification; Trustworthy AI

Education

B.Sc. in Computer Science and Engineering **11 Aug. 2026**
Rajshahi University of Engineering & Technology (RUET) *Rajshahi, Bangladesh*
CGPA: 3.77/4.00 | **Program:** BAETE-accredited CSE program

Research Experience

Undergraduate Thesis: Designing and Evaluating Explanation Interfaces for AI-Assisted Medical Misinformation Detection
Transformer-Based NLP | Model Reliability | Medical Misinformation | Human–AI Evaluation

- Developed a **RoBERTa-based misinformation detection system** using **17,611 multi-source samples**. Applied probability calibration and conformal selective prediction to handle uncertainty; the system automatically classified **91.56%** of cases with **0.9806 macro-F1** on the automatically handled subset, referring uncertain cases for human review.
- Evaluated **model reliability beyond classification accuracy**, studying confidence estimates and selective prediction. Compared content-focused text representations with metadata-enriched inputs to identify potential shortcut learning in credibility classification.
- Built four evidence- and uncertainty-oriented explanation interfaces with **Next.js and FastAPI**. Conducted a controlled study of **60 participants and 1,200 judgment trials**; the most informative interface improved decision accuracy from **73.7% to 90.0%** and reduced reliance on incorrect AI advice from **77.8% to 37.8%**.

Publications

Nahid Montasir Rifat, Tasfia Jannat Tasfi, Foyez Ahmed Dewan, and Barshon Sen. *Evaluating Creative Reasoning and Lateral Thinking in Instruction-Tuned LLMs for Bengali.* 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), 2026.
DOI: https://doi.org/10.1109/QPAIN69676.2026.11546604

Evaluated instruction-tuned LLM reasoning with parallel Bangla–English prompts, identifying mistranslation, cultural mismatch, and logical inconsistencies in cross-lingual responses.

Jabunnesa Jahan Sara, Shahriar Rizvi, and Tasfia Jannat Tasfi. *Beyond Artifacts: Semantic Deepfake Detection and Digital Provenance Tracking with Multimodal Large Language Models.* 2025 28th International Conference on Computer and Information Technology (ICCIT), 2025.
DOI: https://doi.org/10.1109/ICCIT68739.2025.11490141

Developed a multimodal pipeline for semantic deepfake analysis and web-based source verification, emphasizing information credibility and digital provenance beyond visual artifacts.

Selected Achievements

- **1st Place, UIHP 5th Cohort (WayMakers)** — startup innovation, market validation, and MVP development.
- **Champion, Huawei Seeds for the Future Tech-Business Competition**, 2024.
- **International Technology Competition, China** — Participated in a technology competition in China, gaining international exposure to innovation and cross-cultural perspectives.
  Media coverage: The Daily Star — Seeds for the Future Regional Round, China (thedailystar.net)
- **Champion, Hult Prize at RUET** — Startup Innovation, 2023.
- **Government General Scholarship** — all four undergraduate years.
- **Finalist, International Youth Math Challenge** — Top 20% globally, 2024.
- **70% Merit Scholarship**, ISCEA PTAK Prize, 2024.
- **Government Talent Pool Scholarship** — academic excellence in high school.

Selected AI & NLP Projects

BongoSeek — LLM-Powered Conversational Search. Developed a conversational search prototype with language models, context-aware queries, and dialogue memory for multi-turn information retrieval.
Project: github.com/Shion115/Bongo_Seek

EarlyMind — AI-Assisted Child Development Interface. Designed an interactive child-development assessment prototype with user-facing dashboards and accessible, cognitive-load-aware feedback.
Project: github.com/Shion115/EarlyMind

Professional & Leadership Experience

Co-Founder, Delibro **Nov. 2024 – Present**
Social Computing Platform for Community-Based Delivery

- Co-founded a peer-to-peer delivery platform connecting customers with travelers. Contributed to interface design and prototyping around user trust, coordination, safety, and feedback; received support through the **World Bank-supported University Innovation Hub**.

Official Ambassador **Jun. 2025 – Present**
International Computer Science Competition

- Organized programming outreach, mentorship activities, and student community initiatives. **Profile:** icscompetition.org/en/ttasfi

Industrial Trainee **Mar. 2025**
BRAC IT

- Worked on front-end interface design and web-development workflows.

App Developer **Jun. 2022 – Jan. 2023**
bdapps

- Developed and deployed PHP-based applications involving interface design, backend integration, and deployment.

Research & Technical Skills

NLP / LLMs	Python, RoBERTa, Hugging Face Transformers, scikit-learn, Cross-Lingual Error Analysis, LLM Reasoning Evaluation
Trustworthy AI	Probability Calibration, Conformal Selective Prediction, Misinformation Detection, Error Analysis, Multimodal Credibility Verification
Research Methods	Dataset Preprocessing, Model Benchmarking, Controlled User Studies, Human–AI Reliability Evaluation, pandas, Jupyter Notebook, Google Colab
Development	Next.js, FastAPI, Flask, JavaScript, HTML/CSS, C++, PHP, Git, Figma

References

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